
Section C: Advanced RAG

RAG Week Assignment

C1. Multimodal RAG for a Product Manual

Question

You're building a RAG system for a company's product manuals. These manuals have text instructions along with diagrams and photos showing how to assemble the product. A user asks "how do I attach the side panel," and the answer is shown entirely in a diagram, with no accompanying text.

- In multimodal RAG, both the text and images from the manual would be processed and stored so that the system can retrieve the relevant information from any format.
- The user's question would first be converted into an embedding, and the system would search for related text sections as well as useful diagrams or images. For example, "attach the side panel" could retrieve the diagram showing the exact attachment process.
- The retrieved text and image would then be given to a multimodal AI model, which can understand both formats and generate an answer using the available information.
- If we use only text-based RAG, the important diagram may be ignored because it does not contain useful text. As a result, the system may fail to retrieve the actual answer or give an incomplete response.

C2. Unified Embedding Space vs. Separate Indexes with Fusion

Question

Briefly compare the two approaches to multimodal retrieval covered in lecture: unified embedding space vs separate indexes with fusion. Which one would you pick for the product manual scenario above, and why?

- In a unified embedding space, text and images are converted into embeddings in the same space, so related text and images can be searched together.
- In separate indexes with fusion, text and images are stored separately. The system searches both and then combines the results.
- Unified embedding is comparatively simple because different types of information can be searched together in one place.
- For the product manual, I would choose unified embedding because a text query like "how to attach the side panel" can directly find the related diagram as well as the text.